Divisibility of Reversed Consecutive-Integer Concatenations and an Arbitrary-Permutation Extension

Qichao Wang
Hebei University of Technology, Tianjin, China

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Abstract

For every positive integer n, we show that the concatenation of some n consecutive positive integers in decreasing order is divisible by n. Consequently, OEIS A083468 is positive and OEIS A083469 is well-defined for every positive index. We prove a more general assertion: in any even integer base, *every prescribed permutation* of n consecutive integers admits infinitely many equal-width concatenations divisible by n. The proof is constructive and rests only on congruences and the Chinese remainder theorem. We establish existence, not an explicit formula for the least sequence values.

1 The two sequences

Write $u \,\|_b\, v$ for concatenation of the base-b digit strings of positive integers u, v. The sequence OEIS A083468 gives the smallest positive multiple of n obtained by concatenating n consecutive positive integers in *decreasing* order, with the value zero if no such concatenation exists. The companion sequence OEIS A083469 divides that smallest value by n; its definition does not assign a value in a hypothetical nonexistence case [1, 2]. We prove nonexistence never occurs. The proof is independent of the minimality of the listed sequence terms. It can also be viewed as a corollary of the author's even-base arithmetic-progression divisibility theorem, taking step -1 [3]. The independent proof below exposes a permutation strengthening.

2 Divisibility in every prescribed block order

Theorem 1. *Let $b \geq 2$ be an even integer, $n \geq 1$, and let $\pi : \{1, \ldots, n\} \to \{0, \ldots, n-1\}$ be an arbitrary bijection. There exist infinitely many pairs (d, k) of positive integers such that $k, k+1, \ldots, k+n-1$ all have exactly d digits in base b and*

$$n \;\Big|\; (k + \pi(1)) \,\|_b\, (k + \pi(2)) \,\|_b \cdots \|_b\, (k + \pi(n)). \tag{1}$$

In particular, there are infinitely many distinct concatenated integers satisfying (1).

Proof. Factor $n = qm$, where q is the largest divisor of n supported on primes dividing b. Thus $\gcd(q, m) = 1$ and $\gcd(m, b) = 1$. Since b is even, m is odd. Choose an arbitrarily large positive integer d with

$$q \mid b^d, \qquad b^d \equiv 1 \pmod{m}, \qquad (b-1)b^{d-1} \geq q + n - 1. \tag{2}$$

Such d exist arbitrarily far out: divisibility by q holds for all sufficiently large d; the second condition holds when d is a multiple of the multiplicative order of b modulo m (and is vacuous when $m = 1$); the inequality follows for large d. The inclusive interval

$$I_d = [b^{d-1},\, b^d - n] \cap \mathbb{Z}$$

contains at least q consecutive integers, by (2). Choose $k \in I_d$ satisfying

$$k + \pi(n) \equiv 0 \pmod{q}. \tag{3}$$

If $q = 1$, this congruence is vacuous. All n of the integers $k, \ldots, k + n - 1$ are positive and have exactly d base-b digits. Put $B = b^d$. The proposed concatenation is the integer

$$C = \sum_{i=1}^{n} \bigl(k + \pi(i)\bigr) B^{n-i}. \tag{4}$$

Since $q \mid B$, the final summand in (4) is the only one that survives modulo q, so (3) gives

$$C \equiv k + \pi(n) \equiv 0 \pmod{q}.$$

On the other hand, $B \equiv 1 \pmod{m}$, hence the permutation drops out of the residue:

$$C \equiv \sum_{i=1}^{n} (k + \pi(i)) = nk + \frac{n(n-1)}{2} \equiv 0 \pmod{m}.$$

The last congruence holds because $m \mid n$ and m is odd. As $\gcd(q, m) = 1$, the Chinese remainder theorem implies $n \mid C$. There are arbitrarily large admissible d, and the resulting concatenations have increasing lengths nd, proving infinitude. □

Corollary 1. *For every positive integer n, there exist infinitely many concatenations of n consecutive positive integers in decreasing order that are divisible by n. Consequently*

$$\boxed{\text{A083468}(n) > 0 \quad \textit{for every } n \geq 1,}$$

while $\text{A083469}(n) = \text{A083468}(n)/n$ *is well-defined and positive for every* $n \geq 1$.

Proof. In Theorem 1 take $b = 10$ and $\pi(i) = n - i$, so that the blocks occur in descending order. □

3 A concrete witness construction

For the decimal case, factor

$$n = qm, \qquad q = 2^{v_2(n)} 5^{v_5(n)}, \qquad \gcd(m, 10) = 1.$$

Take an arbitrarily large d satisfying

$$q \mid 10^d, \qquad 10^d \equiv 1 \pmod{m}, \qquad 9 \cdot 10^{d-1} \geq q + n - 1.$$

The smallest multiple of q not below 10^{d-1},

$$a = q \left\lceil \frac{10^{d-1}}{q} \right\rceil,$$

lies in $[10^{d-1}, 10^d - n]$. Therefore the integer

$$(a+n-1) \,\|_{10}\, (a+n-2) \,\|_{10} \cdots \|_{10}\, a$$

is divisible by n. For instance, with $n = 4$, $d = 2$ and $a = 12$, we obtain 15141312, which is divisible by 4; this is not the minimum, which is 5432 [1].

Remark 1. *The hypothesis that b is even ensures that m is odd, which is used to annihilate $n(n-1)/2$ modulo m. No such theorem for arbitrary odd bases is claimed here.*

4 Computational checks and scope

The accompanying script `verify_reverse_concatenation.py` constructs witnesses for $1 \le n \le 512$, verifies divisibility using modular concatenation, and independently reproduces the first sixteen A083468 minima. Finite checks are supplementary: the theorem is an unconditional proof for all indices. The proof establishes *existence* and provides an explicit family of witnesses. It does not evaluate the minimal concatenation or its quotient in closed form. The theorem includes the previously observed reverse-progression consequence of [3]; no claim is made that the arbitrary-permutation strengthening is new to the literature.

References

[1] OEIS Foundation Inc., *A083468*, https://oeis.org/A083468 (accessed October 16, 2026).

[2] OEIS Foundation Inc., *A083469*, https://oeis.org/A083469 (accessed October 16, 2026).

[3] Q. Wang, *Divisibility of Concatenated Arithmetic Progressions and Champernowne Prefixes*, https://github.com/dongxuelian2/oeis-papers/tree/main/papers/A075000-A075002, 2026. See the even-base AP divisibility theorem.

[4] F. Luca and B. Teguia Tabuguia, *Concatenations of terms of an arithmetic progression*, Journal of Difference Equations and Applications **32** (2026), 437–456, https://doi.org/10.1080/10236198.2025.2603385.